

Activity: Database Modelling and Implementation for VideoWall System

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Discipline: CEAPI - Database

Tools: www.freesqldatabase.com and MySQL Workbench

Theme: Content Management for VideoWall System

General Objective:

Empower students in the conceptual, logical and physical modelling of a relational database, using MySQL Workbench for design and www.freesqldatabase.com as a DBMS environment for implementation and testing.

Contextualisation:

A VideoWall system is a dynamic visual communication tool, used to display various information in real time, such as news, institutional announcements, course and event programming, restaurant menu, among others. For this system to work efficiently, a well-structured database is crucial to store and manage all the content to be displayed.

Activity Description:

Students must design and implement a database to manage the content of a VideoWall system. The system must be able to store and organise information of different natures.

Minimum Functional Requirements of the VideoWall System (to guide modelling):

Content Registration:

Allow the registration of different types of content (News, Institutional Information, Course Schedule, Event, Restaurant Menu Item).

Each content must have a title, description, creation date and expiration date (for automatic expiration).

Content Categorisation:

Allow the creation of categories to organise the content (e.g. "General News", "HR Announcements", "IT Courses", "Academic Events", "Main Courses").

A content can belong to one or more categories.

Exhibition Scheduling:

Allow scheduling when a specific content (or a content category) should be displayed on a given VideoWall (or specific screen).

It should be possible to set the start and end date and time of the display.

VideoWalls/Screens Management:

Allow the registration of the different screens or VideoWalls available, with an identification and location (e.g. "Main Reception VideoWall", "Block A Corridor Screen").

Media Type Management:

The content can be associated with different types of media (text, image, video). The database must store the path or URL of the media.

Content Prioritisation:

Allow to define a priority level for the contents, so that the most important ones can be highlighted.

Activity Steps:

Part 1: Database Modelling (Using MySQL Workbench)

Group Discussion and Requirements Survey:

Analyse the functional requirements described above.

Identify the main entities involved in the system.

Define the attributes for each entity, including their data types and restrictions (primary, foreign, not null keys, etc.).

Identify the relationships between the entities and their cardinalities.

Creation of the Conceptual Model (MER/DER):

Draw the Entity-Relationship Diagram (DER) using the MySQL Workbench modelling tool.

Logical Model Creation:

Convert the conceptual model to the logical model, normalising the tables (it is suggested up to the 3^a Normal Form).

Creation of the Physical Model:

Generate the SQL script (DDL - Data Definition Language) for the creation of tables, indexes and relationships from the logical model in MySQL Workbench.

Part 2: Database Creation and Management at www.freesqldatabase.com

Account and Database Creation:

Access the website www.freesqldatabase.com

Each student (or group) must create an account and request the creation of a MySQL database. Write down the access credentials (hostname, bank name, user and password).

Attention: www.freesqldatabase.com is a test service and may have instabilities or be deleted. Save the SQL scripts locally!

Connection via MySQL Workbench:

Set up a new connection in MySQL Workbench using the credentials obtained from www.freesqldatabase.com.

Test the connection.

SQL Script Execution (DDL):

Open the SQL script generated in Part 1 in MySQL Workbench.

Run the script in the database created at www.freesqldatabase.com to create the table structure.

Verify that all tables and their relationships have been created correctly.

Database Stocking (DML - Data Manipulation Language):

Create SQL scripts with INSERT commands to populate the tables with sample data. The data must be consistent with the purpose of each table and allow you to test the system's functionalities. It is suggested to register:

At least 3 VideoWalls/Screens.

At least 5 Categories.

At least 10 Contents of various types (news, events, menu items, etc.), associating them with categories and media.

At least 5 Exhibition Schedules, linking content to VideoWalls.

Query Creation (DQL - Data Query Language):

Prepare and execute at least 5 SQL queries that demonstrate the usefulness of the database. Examples:

List all active content (that has not expired) for a given category.

List the content scheduled to be displayed on a specific VideoWall today.

Show all events scheduled for next month.

Display the restaurant menu for today.

List the latest news (last 7 days) with high priority.

Deliverable:

Modelling Document:

Brief description of the project.

Entity-Relationship Diagram (DER).

Logical Model (description of tables, attributes, data types, primary and foreign keys).

SQL scripts:

DDL script for creating the database structure.

DML script for database settlement with sample data.

DQL scripts with the elaborated queries.

Brief Presentation/Demonstration:

Students have to create a presentation SWAY that shows the structure of the database, the tables created in db4free.net (via MySQL Workbench) and the execution of some queries.

Entity Suggestions (Starting Point):

VideoWalls (or Screens)

Contents

TypesContent (News, Event, Menu, etc.)

Categories

ContentCategories (Associative table for N:M between Contents and Categories)

Media (associated with content)

Schedules Exhibition

Users (to manage who registers the content - optional, can add complexity)

Evaluation Criteria:

Modelling: Correctness of the DER and the logical model, normalisation, adequate definition of attributes and relationships.

Implementation: Correct creation of tables in db4free.net, use of keys and restrictions.

Settlement: Quality and consistency of sample data.

SQL queries: Complexity, relevance and correctness of the elaborated queries.

Organisation and Documentation: Clarity of deliverables.

Important Observations for Students:

The db4free.net service is for testing. Do not store sensitive or critical data.

Make regular backups of your SQL scripts. The database on db4free.net can be deleted without prior notice.

Performance on db4free.net may vary.

Work as a team to discuss the best modelling, but the implementation of scripts and queries can be individualised if you prefer.

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